MIL-I-55510B(EL)
30 March 1973
SUPERSEDING
MIL-I-55510A(EL)
30 January 1970

MILITARY SPECIFICATION

INDICATOR, CHANNEL ALIGNMENT ID-1189( )/PR

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a type of portable, battery operated alignment indicator designated Indicator, Channel Alignment ID-1189( )/PR, constructed to provide a means to align Receiving Set, Radio AN/PRR-9( ) and Transmitting Set, Radio AN/PRT-4( ) on any desired channel in the operating frequency range of 47.00 to 57.00 megahertz (MHz). It also provides a means for testing Battery, Dry BA-505/U and Battery, Dry BA-399/U respectively for the radio sets (See 6.1 and 6.6)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallet, Materials - Handling, Wood.
PPP-B-585	Boxes, Wood Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-F-320	Fiberboard; Sheet, Stock and Cut Shapes.
PPP-T-76	Tape; Pressure Sensitive Adhesive Paper, Water Resistant for Carton Sealing.
PPP-T-97	Tape; Pressure-Sensitive Adhesive, Filament-Reinforced.
QQ-S-781	Steel Strapping, Flat.

MILITARY

MIL-P-116	Preservation, Methods of.

FSC 5820

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MILITARY (Continued)

MIL-P-11268	Parts, Materials, and Processes used in Electronic Equipment.
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment.

STANDARDS

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized and Containerized Unit Loads.
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-461	Electromagnetic Interference Characteristics Requirement for Equipment.
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	Environmental Test Methods.

DRAWINGS

ELECTRONICS COMMAND

SC-DL-523600	Indicator, Channel Alignment ID-1189( )/PR

(Copies of specification, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First Article.- When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3 and 6.2.

3.2 Construction.- The alignment indicator shall be constructed in accordance with the following Drawing and Data List: SC-DL-523600.

3.3 Parts, Materials, and Processes; General.- In addition to the requirements of this specification, the requirement of MIL-P-11268, including the selection requirements therein, shall apply.

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3.4 Cleaning.-

3.4.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, cases, etc.

3.4.2 Units.- After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.5 Controls.- All movable controls shall operate properly without binding or other undue restriction. Controls shall not be assembled in a misaligned condition.

3.6 Finish.- The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (See 4.4) The final paint film on Type I surfaces shall be green color (Olive Drab), semigloss enamel matching a color chip provided by the procuring agency. (See 6.3)

3.7 Marking.-

3.7.1 General.- Marking shall conform to Specification MIL-M-13231. (See 4.4) Case markings shall be Group I as described in that specification.

3.7.2 Visibility.- Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.7.3 Serial Numbers.- Indicator, Channel Alignment ID-1189( )/PR shall be serial numbered.

3.8 Securing of Parts.-

3.8.1 General.- Brackets, lugs, flanges, inserts, bolts and other mounting arrangements shall retain items securely when subjected to specified service conditions.

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3.8.2 Securing of Parts by Threaded Fasteners.- Nylok or other screws with plastic devices, and loctite or similar sealouts shall not be subjected to temperatures in excess of 250°F. Manufacturers' instructions shall be followed in the application of sealouts.

3.9 Service Conditions.- The equipment shall meet the following service conditions.

3.9.1 Temperature (See 4.7.1).- The equipment shall meet the following requirements when subjected to temperature test procedures as shown in Methods 501 and 502 of MIL-STD-810.

a. Low.- During testing as specified in 4.7.1.1, the equipment shall meet the following requirements:

Paragraph	Measurements	Performance
3.10.1	Battery, BA-399/U (Test for)	4.9.1
3.10.2	Battery, BA-505/U (Test for)	4.9.2
3.10.3	Internal battery	4.9.3
3.10.4	DCVM sensitivity	4.9.4
3.10.5	Transmitter current	4.9.5
3.10.6	Field strength	4.9.6
3.10.7	Receiver oscillator current	4.9.7
3.10.8	Regulated voltage	4.9.8
3.10.9	Receiver audio stage current	4.9.9
3.10.10	Signal generator	4.9.10
3.10.12a & b	Frequency setability	4.9.12
3.10.13	Frequency stability	4.9.13a & b

After testing, the equipment shall meet full specification performance for these measurements.

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b. High.- During testing as specified in 4.7.1.2, the equipment shall meet the following requirements:

Paragraph	Measurements	Performance
3.10.1	Battery, BA-399/U (Test for)	4.9.1
3.10.2	Battery, BA-505/U (Test for)	4.9.2
3.10.3	Internal battery	4.9.3
3.10.4	DCVM sensitivity	4.9.4
3.10.5	Transmitter current	4.9.5
3.10.6	Field strength	4.9.6
3.10.7	Receiver oscillator current	4.9.7
3.10.8	Regulated voltage	4.9.8
3.10.9	Receiver audio stage current	4.9.9
3.10.10	Signal generator	4.9.10
3.10.12a & b	Frequency setability	4.9.12
3.10.13	Frequency stability	4.9.13a & b

After testing, the equipment shall meet full specification performance for these measurements.

3.9.2 Humidity (Moisture-Resistance).- During testing as specified in 4.7.2, the equipment shall meet the following requirements:

Paragraph	Measurements	Performance
3.10.3	Internal battery	4.9.3
3.10.4	DCVM sensitivity	4.9.4
3.10.7	Receiver oscillator	4.9.7
3.10.10	Signal generator	4.9.10
3.10.11	Receiver test signal	4.9.11

After testing, the equipment shall meet full specification performance for these measurements.

3.9.3 Altitude (Elevation).- The equipment shall meet full specification performance for the following, both during and after testing in accordance with paragraph 4.7.3:

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Paragraph	Measurements	Performance
3.10.4	DCVM sensitivity	4.9.4
3.10.10	Signal generator	4.9.10

3.9.4 Leakage (Immersion).- The equipment shall show no evidence of leakage after it is tested as specified in 4.7.4.

3.9.5 Orientation.- When tested as specified in 4.7.5, the equipment shall meet full specification performance for the following measurements:

Paragraph	Measurements	Performance
3.15	Operational inspection	4.11

3.9.6 Dust.- After testing as specified in 4.7.6, the equipment shall meet full specification performance for the following measurements:

Paragraph	Measurements	Performance
3.15	Operational inspection	4.11

3.9.7 Vibration, Bounce and Shock.-

3.9.7.1 Vibration, Internal.- The amplitude of vibration of any part, subassembly or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 Hertz. After being tested in accordance with paragraph 4.7.7, there shall be no physical damage and the equipment shall meet the performance requirements of paragraphs 4.9.3 and 4.9.10.

3.9.7.2 Bounce and Shock.- The equipment shall be capable of meeting the requirements of Table I.

TABLE I - Bounce and Shock

Test	Paragraph No.	Performance after test
Bounce	4.7.8	Specified performance (Note A) No physical damage.
Shock, Bench-Handling	4.7.9b	Specified performance (Note A) No physical damage.
Shock-Drop	4.7.9a	Operable. (Note A) Any physical damage shall be minor only and shall not affect the operational capability of the equipment.

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Note A: The channel alignment indicator shall meet specified performance for the following requirements: 3.10.1, 3.10.3, 3.10.7, 3.10.10, and 3.10.12 (test to be in accordance with paragraph 3, General Requirements of MIL-STD-810).

3.9.8 Fungus.- The equipment (including accessories) shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.16. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.9.9 Salt Fog.- After testing as specified in 4.17, the equipment shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish of metal surface.

3.10 Electrical Characteristics.-

3.10.1 Meter Test for Battery, Dry BA-399/U (See 4.9.1).- A 12.0 volt d.c. voltage impressed at the XMTR BAT connector shall produce an alignment indicator meter indication at the upper edge of the yellow area. The voltage required for this reading shall be between 11.76 volts and 12.24 volts inclusive.

3.10.2 Meter Test for Battery, Dry BA-505/U (See 4.9.2).- A 12.0 volt d.c. voltage impressed at the RCVR BAT connector shall produce a meter indication at the upper edge of the yellow area. The voltage required for this reading shall be between 4.9 volts and 5.1 volts inclusive.

3.10.3 Meter Test for Internal Battery (See 4.9.3).- A 12.0 volts d.c. voltage impressed at the internal battery plug shall produce a meter indication at the lower edge of the yellow area. The voltage required for this reading shall be between 9.8 volts and 10.2 volts inclusive.

3.10.4 Meter Test for DCVM Sensitivity (See 4.9.4).- The voltmeter shall indicate plus and minus 1.25 volts to 1.4 volts full scale when zero centered.

3.10.5 Meter Test for Transmitter Current (See 4.9.5).- Test for transmitter RF alignment, when transmitter current is monitored by the alignment indicator. Transmitter current for full scale meter indication shall be between 190 milliamperes and 210 milliamperes inclusive.

3.10.6 Field Strength Monitor (See 4.9.6).- A 200 millivolt signal impressed at the XMTR ANT shall produce an alignment indicator meter indication between 30% full scale and 70% full scale.

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3.10.7 Receiver Oscillator Current (See 4.9.7).- Test for receiver oscillator current (monitored for receiver oscillator adjustment) shall produce an alignment indicator meter indication between 40% full scale and 60% full scale.

3.10.8 Regulated Voltage (See 4.9.8).- The range for receiver regulated d.c. supply voltage shall not be less than 5.6 volts and not more than 6.51 volts.

3.10.9 Receiver Audio Stage Current (See 4.9.9).- Test for receiver audio stage current (monitored for receiver squelch adjustment) shall produce an alignment indicator meter indication between 50% full scale and full scale.

3.10.10 Signal Generator Output (See 4.9.10).- The output of the alignment indicator signal generator measured at the antenna socket shall be between 500 microvolts and 1000 microvolts.

3.10.11 Receiver Test Signal (See 4.9.11).- The RCVR TEST SIG control shall decrease the test signal from the maximum signal measured with RCVR TEST SIG control in extreme clockwise of Step 6 position in increments as follows:

Step 5	9.5 db ± 1 db
Step 4	9.5 db ± 1 db
Step 3	9.5 db ± 1 db
Step 2	9.5 db ± 1 db
Step 1	5.0 db minimum

3.10.12 Frequency Setability (See 4.9.12).-

a. The frequency setability of the alignment indicator signal generator shall be within plus or minus 200 KHz of the desired frequency at room temperature.

b. At temperature extremes of -40°F, and +140°F, the frequency setability shall be within plus or minus 250 KHz of the desired frequency.

3.10.13 Frequency Stability (See 4.9.13).- The frequency of the signal generator shall not drift over 10 KHz during any 10 minute period at any ambient temperature between -40°F and +140°F.

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3.10.14 Input Voltage Range (See 4.9.14).- The channel alignment indicator shall meet full specification requirements when operating from a d.c. power source range of 10.0 to 16.5 volts.

3.10.15 Oscillator Operation (See 4.9.15).- The oscillator in the signal generator shall start with an input voltage of 9.0 volts d.c.

3.11 500 Hour Operational Requirement (See 4.10).- The channel alignment indicator shall be operated continuously for a period of 500 hours under the conditions specified in 4.10 with no replacement of parts or readjustment. At the end of the 500 hour period, the channel alignment indicator shall meet the tests specified in 4.10.2.

3.12 Electromagnetic Interference Characteristics.- The equipment shall meet the electromagnetic interference requirements specified in MIL-STD-461 for class 1B. Tests and test methods shall be as specified in paragraph 4.18. If the contractor considers it desirable to modify the equipment proper in order to facilitate conformance to the specified test limits, he shall obtain approval from the contracting officer before proceeding with the modification.

3.13 Preconditioning.-

3.13.1 Bounce Preconditioning.- The channel alignment indicator shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.1.

3.13.2 Burn-In Debugging.- Each channel alignment indicator on order shall be subjected to a burn-in operation of 46 hours at 12.0 volts d.c. input. The last 22 hours of the 46 hour burn-in shall be failure free. Indicator, channel alignment shall meet the requirements specified herein, without subsequent processing after subject to paragraph 4.6.2.

3.14 Air-Seal Test, Provision for.- The transit case, or the equipment inclosure when no transit case is employed, shall provide means for enabling performance of the air-seal test. (See 4.15)

3.15 Operational Inspection.- The operational test specified in 4.11 shall be performed to determine that the equipment is operable prior to packaging and at such other times as required.

3.16 Technical Literature.- Technical literature shall be furnished as specified in the contract. (See 6.2)

3.17 Workmanship.- The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of paragraphs 3.2 through 3.8 herein.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.- Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of Inspection.- Inspection shall be classified as follows:

a. First Article inspection (does not include preparation for delivery). (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection:

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.12)

4.3 First Article.- Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First Article Units.- The contractor shall furnish five first article units of the complete Indicator, Channel Alignment ID-1189/PR.

4.3.2 First Article Inspection.- This inspection shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B, and Group C (see Tables III, IV, and V, respectively). The inspection shall be performed in the following order: 4.4, 4.6, Group A and Group B for all units; and Group C as specified in Table II.

4.3.3 First Article Data.- The first article test plan and test report(s) shall be as required in the contract or purchase order.

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TABLE II - First Article Environmental Inspection

Inspection	Number of Samples				
	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5
Non-Damaging					
Temperature					
Low	1				
High	2				
Orientation	3				
Altitude (Elevation)		1			
Leakage (Immersion)		2			
Dust		3			
500 Hour Operational			1		
EMI					1
Potential Damaging					
Humidity (Moisture-Resistance)	4				
Vibration		4			
Bounce		5			
Shock, Bench-Handling		6			
Shock, Drop		7			
Fungus (Note 3)				1	
Salt Fog (Note 3)					2

Note 1: Other First Article inspection may precede, follow, be interspersed between the non-damaging tests.

Note 2: The order on non-damaging tests and the choice of which units are subjected to the non-damaging tests may be varied if convenient, except that the vibration test should be performed on the same unit used for the bounce and shock tests.

Note 3: The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

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4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where Required
Parts, material & process used in Electronic Equipment	3.3
Finish, protective	3.6
Marking	3.7
Workmanship	3.17

4.5 Quality conformance inspection of units before preparation for delivery.- Each unit produced under this specification including First Articles shall be first subjected to bounce preconditioning (see 3.13.1), and then to the 46 hour burn-in debugging (see 3.13.2) prior to contractor's Group A inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.3.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the government, at its discretion, may perform all or any part of the specified inspection to verify the contractor's compliance with specified requirements (see 6.7). Test equipment for government verification inspection shall be made available by the contractor.

4.5.1 Group A Inspection.- Each unit on contract or purchase order shall be inspected for conformance to the inspection specified in Table III. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table III.

4.5.1.1 Order of Inspection with Group A.- Group A inspection shall be performed in an order satisfactory to the government, except that the operational in 4.11 shall be next to last and the air seal in 4.15 shall be last. (See 6.5)

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TABLE III - Group A Inspection

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Visual and Mechanical				
Printed Circuit Boards	3.17	4.14		
A-1 Functional	3.17	4.14	0.65%	2.5%
A-2 Attenuator	3.17	4.14	0.65%	2.5%
A-3 VFO	3.17	4.14	0.65%	2.5%
Case CY-6078	3.17	4.14	1%	4%
Test Probe Assembly	3.17	4.14	1%	4%
Test Probe Test	3.17	4.14	1%	4%
Antenna	3.17	4.14	1%	4%
Screwdriver	3.17	4.14	1%	4%
Tool Alignment	3.17	4.14	1%	4%
Card Receiver	3.17	4.14	1%	4%
Card Transmitter	3.17	4.14	1%	4%
Plate Assembly	3.17	4.14	1%	4%
Connector	3.17	4.14	1%	4%
Antenna Plug	3.17	4.14	1%	4%
ID-1189 (without case)	3.17	4.14	1%	4%
ID-1189 (with case)	3.17	4.14	1%	4%
Insert Radio AN/PRR-9 chassis in Receiver Alignment Receptacle	3.17	4.14	1%	4%
Insert Transmitter AN/PRT-4 chassis in Transmitter Alignment Receptacle	3.17	4.14	1%	4%
Insert Battery BA-399/U in Transmitter Battery Receptacle	3.17	4.14	1%	4%
Insert Battery BA-505/U in Receiver Battery Receptacle				
Electrical				
Battery, BA-399/U (Test for)	3.10.1	4.9.1	1%	NA
Battery, BA-505/U (Test for)	3.10.2	4.9.2		
Internal Battery	3.10.3	4.9.3		
DCVM Sensitivity	3.10.4	4.9.4		
Transmitter Current	3.10.5	4.9.5		
Field Strength	3.10.6	4.9.6		
Receiver Oscillator	3.10.7	4.9.7		
Voltage Regulation	3.10.8	4.9.8		
Receiver Audio Stage Current	3.10.9	4.9.9		
Signal Generator	3.10.10	4.9.10		
Receiver Test Signal	3.10.11	4.9.11		
Frequency Setability	3.10.12a	4.9.12		
Frequency Stability	3.10.13	4.9.13a		
Oscillator Operation	3.10.15	4.9.15		

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TABLE III - Group A Inspection

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Electrical (cont'd)				
Air Seal	3.14	4.15	1%	4%
Operational	3.15	4.11	1%	

NOTE: All electrical defects are considered major.

4.5.2 Group B Inspection.- This inspection (including sampling) shall conform to Table IV and to the special inspection levels of Table I of Standard MIL-STD-105. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met Group A inspection.

TABLE IV - Group B Inspection

Inspection	Req Para	Insp Para	AQL
Relief Valve	SM-B-523692		4.0%
Immersion	3.9.4	4.7.4	6.5%
Input Voltage Range	3.10.14	4.9.14	4.0%
Signal Generator Output	3.10.10	4.9.10	

4.5.2.1 Group B Sampling Plans.- The Group B sampling plans, for the AQL's listed in Table IV shall be as follows:

AQL	Inspection Level
4.0%	S-4
6.5%	S-4

4.5.2.2 Order of Inspection within Group B.- Group B inspection shall be performed in any order which is satisfactory to the government.

4.5.3 Group C Inspection.- Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table V. Samples shall be selected in accordance with 4.5.3.1.

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4.5.3.1 Sampling for Inspection of Equipment.- Units selected for each Group C inspection shall be selected without regard to their quality in accordance with the following:

TABLE V - Group C Inspection

Inspection	Req Para	Insp Para
Subgroup I		
Temperature	3.9.1	4.7.1
Humidity (moisture-resistance)	3.9.2	4.7.2
Vibration	3.9.7	4.7.7
Bounce	3.9.7	4.7.8
Shock, Drop	3.9.7	4.7.9a
Subgroup II		
Altitude (elevation)	3.9.3	4.7.3
Orientation	3.9.5	4.7.5
Dust	3.9.6	4.7.6
Shock, Bench-Handling	3.9.7	4.7.9b
500 Hour Test	3.11	4.10
Subgroup III		
Antenna Assembly	SM-C-523588	
50 lb Pull Test		
25 lb Pull Test		
Bend Test		
Subgroup IV		
*Fungus	3.9.8	4.16
*Salt Fog	3.9.9	4.17
Electromagnetic Interference Characteristics	3.12	4.18

*These tests shall be performed on a separate sample.

4.5.3.1.1 Subgroup I.- For this subgroup, two units from the first 50 production units shall be selected for this first Group C inspection. For subsequent Group C inspections, two units from each successive 100, or two per month if less than 100 are produced monthly, shall be selected.

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4.5.3.1.2 Subgroup II.- For this subgroup, two units from the first 50 production units shall be selected for the first Group C inspection. For subsequent Group C inspections, two units shall be selected from each successive 500, or two every two months, if less than 500 are produced during this two month period.

4.5.3.1.3 Subgroup III.- For this subgroup, two units from the first 50 production units shall be selected for this Group C inspection. For subsequent Group C inspections, two units from each successive 250, or two per month, if less than 250 are produced monthly, shall be selected.

4.5.3.1.4 Subgroup IV.- For this subgroup, two units from the first 50 production units shall be selected for the first Group C inspection. No subsequent Group C inspection for the tests in this subgroup shall be required.

4.5.3.2 Non-compliance.- The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (1) the process and materials, as applicable, and (2) all units of product which were manufactured under the same conditions and which the government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective. (See 6.4)

4.5.4 Reinspection of Conforming Group C Sample Units.- Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A and B inspection.

4.5.5 Disposition of Group C Sample Units.- Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the subject contract or order.

4.6 Preconditioning (See 3.13).-

4.6.1 Bounce Preconditioning.- The channel alignment indicator shall be placed in its normal operating position on the table of the package tester, type 1000 SC, as made by the L. A. B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have

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a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the equipment to a distance of three inches as the equipment bounces. The equipment shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, adjusted, cleaned, or otherwise changed prior to acceptance inspection. (See 3.13.1)

4.6.2 Burn-In Debugging (See 3.13.2).-

4.6.2.1 Test Conditions.- The Channel Alignment Indicator ID-1189/PR shall be continuously operated for a period of 46 hours at a temperature of 40°C ± 5° (95°F to 113°F) and with an input voltage of 12.0 volts d.c. While the unit is in this operating mode, vibration shall be performed as follows: 2.2 G ± 10% at any nonresonant frequency between 20 and 60 Hz measured at the mounting points of the equipment. This vibration shall be applied for 10 minutes during each hour of equipment operating time.

4.6.2.2 Failures.- In the event of a failure during burn-in debugging, the contractor will perform all necessary rework (including replacement of components, modules and wiring, etc.) to correct the failure cause. If the failure occurs during the first 24 hours of the burn-in period, the unit will be reworked and returned to burn-in. If the failure occurs during the last 22 hours of the burn-in period, the unit will be reworked and resubmitted to burn-in until it successfully survives the 22 hour failure free period. If after successful completion of burn-in debugging, the equipment fails either the Group A or Group B tests, the equipment will be reworked and resubmitted to burn-in debugging.

4.6.2.3 Compliance.- Units passing burn-in debugging will be submitted to Group A, Group B, and Group C inspection, in that order, without benefit of repair, cleaning, realignment, or other changes, unless otherwise specified herein.

4.7 Service Conditions Tests (See 3.9).-

4.7.1 Temperature Test (See 3.9.1).-

4.7.1.1 Low Temperature.- The equipment shall be tested in accordance with Method 502, Procedure 1 of MIL-STD-810. The storage temperature shall be -80°F and maintained until temperature stabilization of the test item is reached, not to exceed 24 hours, at any one time. The operating temperature shall be -40°F. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.9.1a.

4.7.1.2 High Temperature.- The equipment shall be tested in accordance with Method 501, Procedure II, of MIL-STD-810. The operating temperature shall be 140°F. Prior to testing, the equipment shall meet full specification performance for the measurement specified in 3.9.1b.

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4.7.2 Humidity (Moisture-Resistance) Test.- The equipment shall be tested in accordance with Method 507, Procedure II, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.9.2. Measurements shall be made during the periods shown on the figure for the cycle. Do not remove equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If any equipment fails to meet the specified performance during cycling, it shall be re-aligned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment still fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test. The channel alignment indicator shall be tested with the case removed.

4.7.3 Altitude (Elevation) Test.- The equipment shall be tested in accordance with Method 500, Procedure I, of MIL-STD-810. When in Step 2 of Procedure I, the case cover shall be closed. In Step 3 of Procedure I, the case cover shall be open. Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.9.3.

4.7.4 Immersion.- The equipment, as prepared for field transportation, shall be tested in accordance with Method 512, Procedure I, of MIL-STD-810. (See 3.9.4)

4.7.5 Orientation Test.- The equipment shall be inclined for a minimum of 5 minutes, in each plane (forward, backward, left, and right) to an angle of 20 +3, -1 degrees. Measurements shall be made during inclination in each plane as specified in 3.9.5.

4.7.6 Dust.- The equipment shall be tested in accordance with Method 510, Procedure I, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.9.6.

4.7.7 Vibration.- The equipment shall be tested in accordance with Method 514.1, Procedure IX, Part 1, of MIL-STD-810. After testing, the equipment must comply with 3.9.7.1.

4.7.8 Bounce Test.- The equipment shall be tested in accordance with Method 514.1, Procedure XI, Part 2, of MIL-STD-810. The equipment shall be energized near the end of each 1/2 hour period, and shall be tested for compliance with 3.9.7.2.

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4.7.9 Shock Test (See 3.9.7.2).-

a. Drop.- The equipment shall be tested in accordance with Method 516.1, Procedure II of MIL-STD-810.

b. Bench-Handling.- The equipment shall be tested in accordance with Method 516.1, Procedure V of MIL-STD-810.

4.8 Standard Test Conditions.- All measurements shall be made under the test conditions listed below unless otherwise specified in the individual tests:

Temperature	With test conditions, para 3, General Requirements, MIL-STD-810.
Humidity	With test conditions, para 3, General Requirements, MIL-STD-810.
Standard receiver load	2-pin adapter plug with 1000 ohm ± 1% and 3900 ohm ± 1% resistors in series between pins (1000 ohm resistor to large pin)
Nominal input voltage	12.0 volts DC
50 ohm ± 1 ohm, 5 watt Transmitter load, resistor (non-inductive type)	4-pin adapter plug with 50 ohm Load resistor (non-inductive type)
Frequency	51.00 MHz

4.9 Electrical Tests.-

4.9.1 Meter Test for Battery, Dry BA-399/U (See 3.10.1).- The test for Battery BA-399/U, shall be performed as specified below. The battery test indication shall be performed to determine compliance with 3.10.1.

Test Procedures.-

a. Apply 12.0 volts d.c. from a d.c. supply to the XMTR BAT plug (upper right terminal is positive polarity, lower left terminal is negative polarity).

b. Set FUNCTION SWITCH to OFF position.

c. Depress BAT TEST switch to XMTR position.

d. Adjust d.c. supply for alignment indicator meter reading at the upper edge of yellow area.

e. Measure the d.c. supply voltage with a d.c. voltmeter.

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f. Note and record the d.c. voltmeter reading to determine compliance with 3.10.1.

4.9.2 Battery, Dry BA-505/U (See 3.10.2).- The test for receiver Battery, BA-505/U, shall be performed as specified below. The battery test indication shall be performed to determine compliance with 3.10.2.

Test Procedures.-

a. Apply 5.0 volts d.c. from a d.c. supply to the RCVR BAT plug (small terminal is negative polarity).

b. Set FUNCTION SWITCH to OFF position.

c. Depress BAT TEST switch to RCVR position.

d. Adjust d.c. supply for alignment indicator meter reading at the upper edge of yellow area.

e. Measure the d.c. supply voltage with a d.c. voltmeter.

f. Note and record the d.c. voltmeter reading to determine compliance with 3.10.2.

4.9.3 Meter Test for Internal Battery (See 3.10.3).- The channel alignment indicator internal battery test indication shall be performed as specified below. The battery test indication shall be performed to determine compliance with 3.10.3.

Test Procedures.-

a. Apply 10.0 volts d.c. from a d.c. supply to the plug in the internal battery compartment (upper left pin is positive polarity, lower right pin is negative polarity).

b. Set FUNCTION SWITCH to INT BAT position.

c. Adjust d.c. supply for alignment indicator meter reading at lower edge of the yellow area.

d. Measure the d.c. supply voltage with a d.c. voltmeter.

e. Note and record the d.c. voltmeter reading to determine compliance with 3.10.3.

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4.9.4 Meter Test for DCVM Sensitivity (See 3.10.4).- The alignment indicator is used as a d.c. voltmeter for the transmitter VCO alignment. The DCVM sensitivity measurement shall be performed in accordance with the following to determine compliance with 3.10.4.

Test Procedures.-

a. Apply 12.0 volts from a d.c. supply to the plug in the internal battery compartment.

b. Set the FUNCTION SWITCH to VCO position, meter shall be centered.

c. Apply a positive voltage from a d.c. supply to the alignment indicator test probe.

d. Adjust the d.c. voltage to the test probe for an indicator meter reading at the extreme right end of the meter scale.

e. Measure the d.c. voltage with a voltmeter.

f. Note and record the voltmeter reading for compliance with 3.10.4.

g. Apply a negative voltage from a d.c. supply to the alignment indicator test probe.

h. Adjust the d.c. voltage to the test probe for an indicator meter reading at the extreme left end of the meter scale.

i. Measure the d.c. voltage with a voltmeter.

j. Note and record the voltmeter reading for compliance with 3.10.4.

4.9.5 Meter Test for Transmitter Current (See 3.10.5).- Test for transmitter current is monitored to indicate proper alignment of the transmitter RF circuits. Test procedure shall be used to determine compliance with 3.10.5.

Test Procedures.-

a. Connect the 50 ohm, 5 watt non-inductive load resistor with adapter plug to the XMTR ALIGN socket on the alignment indicator.

b. Connect a variable d.c. voltage supply in series with a d.c. milliammeter to the XMTR BAT terminals of the alignment indicator (upper right pin is positive polarity, lower left pin is negative polarity).

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c. Set FUNCTION SWITCH to XMTR RF position.

d. Adjust d.c. voltage supply until alignment indicator meter reads to the extreme right end of scale.

e. Note and record milliammeter reading to determine compliance with 3.10.5.

4.9.6 Field Strength (See 3.10.6).- The alignment indicator used as a field strength meter for transmitter antenna loading coil alignment shall meet the requirements of 3.10.6 using the test procedure specified below. This test shall be made at the operating frequency of 51 Megahertz.

Test Procedures.-

a. Supply an unmodulated RF signal of 200 millivolts from the 50 ohm output of a signal generator to a small probe inserted into the alignment indicator XMTR ANT socket.

b. Set the signal generator output frequency to 51 Megahertz.

c. Set the alignment indicator FUNCTION SWITCH to XMTR ANT position.

d. Note and record the indicator meter reading to determine compliance with 3.10.6.

4.9.7 Receiver Oscillator Current (See 3.10.7).- Test for the receiver oscillator current is used as a means to set the proper oscillator frequency. Test procedure shall be used to determine compliance with 3.10.7.

Test Procedures.-

a. Apply 12.0 volts from a d.c. supply to the plug in the alignment indicator internal battery compartment (upper left pin is positive polarity, lower right pin is negative polarity).

b. Use a standard receiver load adapter, consisting of two series resistors, one 1000 ohm, 1/4 watt resistor and one 3900 ohm, 1/4 watt resistor, connected between the two pins of the receiver adapter plug. The 1000 ohm resistor shall be connected to the large pin.

c. Plug the standard adapter load into the receiver connector assembly block.

d. Set the alignment indicator FUNCTION SWITCH to OSC position.

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e. Place the alignment indicator test probe tip at the junction of the two series standard load resistors (between the 1000 ohm and 3900 ohm resistors).

f. Note and record the alignment indicator meter reading for compliance with 3.10.7.

4.9.8 Regulated Voltage (See 3.10.8).- The range of receiver d.c. regulated voltage supply measurements shall be performed when the regulated voltage output is properly loaded with 12 ma. The regulated voltage is measured as specified in test procedure below. The regulated voltage is measured and recorded with input voltages of 10 VDC and 16.5 VDC. The recorded results shall comply with 3.10.8.

Test Procedures.-

a. Connect a variable standard d.c. voltage supply to the plug in the alignment indicator internal battery compartment (upper left pin is positive polarity, lower right pin is negative polarity).

b. Plug into the receiver connector assembly block a standard receiver load adapter, consisting of two series resistors, one 1000 ohm, 1/4 watt resistor and one 3900 ohm, 1/4 watt resistor connected between the two pins of the receiver adapter plug.

c. Set the FUNCTION SWITCH to OSC position.

d. The regulated voltage shall be measured by connecting a DC VTVM across the two resistor standard receiver load. The large pin of the adapter plug is positive polarity.

4.9.9 Receiver Audio Stage Current (See 3.10.9).- Test for the receiver audio stage current is measured to indicate the proper adjustment of the receiver squelch. Test procedures below shall be used to indicate compliance with 3.10.9.

Test Procedures.-

a. Adjust the d.c. voltage supply for an output voltage of 1.8 volts.

b. Connect the positive lead from the d.c. supply to the tip of the RCVR OUTPUT plug.

c. Connect the negative lead from the power supply to the alignment indicator chassis.

d. Set the alignment indicator FUNCTION SWITCH to SQUELCH position.

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e. Note and record the reading of the alignment indicator meter for compliance with 3.10.9.

4.9.10 Signal Generator Output (See 3.10.10).- The alignment indicator signal generator output measurements shall be performed as specified below. The signal output shall be measured at 47.0, 51.0, and 57.0 Megahertz.

Test Procedures.-

a. Apply 12.0 volts from a d.c. supply to the plug in the alignment indicator internal battery compartment (upper left pin is positive polarity, lower right pin is negative polarity).

b. Set the FUNCTION SWITCH to RCVR ANT position.

c. Set the FREQUENCY MC control to the desired frequency.

d. Connect the Noise and Field Intensity Meter to the ANTENNA socket on the receiver antenna block of the alignment indicator.

e. Note and record the Noise and Field Intensity Meter reading for compliance with 3.10.10.

4.9.11 Receiver Test Signal (See 3.10.11).- The receiver test signal measurements shall be performed as specified below to determine compliance with 3.10.11.

Test Procedures.-

a. Apply 12.0 volts from a d.c. supply to the plug in the alignment indicator internal battery compartment (upper left pin is positive polarity, lower right pin is negative polarity).

b. Set the FUNCTION SWITCH to RCVR ANT position.

c. Set the alignment indicator RCVR TEST SIG control to the extreme clockwise or step 6 position.

d. Connect the Noise and Field Intensity Meter to the ANTENNA socket on the receiver antenna block of the alignment indicator.

e. Measure the signal output as the RCVR TEST SIG control is rotated counterclockwise through each step from step 6 to step 1.

f. Note and record the incremental decrease in signal output at each step for compliance with 3.10.11.

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4.9.12 Frequency Setability (See 3.10.12).- This test shall be performed in accordance with the test procedures specified below. The test results shall comply with 3.10.12.

Test Procedures.-

a. Apply 12.0 volts from a d.c. supply to the plug in the alignment indicator internal battery compartment (upper left pin is positive polarity, lower right pin is negative polarity).

b. Set the FUNCTION SWITCH to COARSE RF position.

c. Connect the electronic counter to the RCVR ANTENNA socket by means of a small probe inserted in the socket.

d. At +77°, +140°, and -40° (± 5°) F, successively set the FREQUENCY MC dial to 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, and 57 Megahertz positions.

e. Note and record the readings of electronic counter for compliance with 3.10.12.

4.9.13 Frequency Stability (See 3.10.13).- These tests shall be performed in accordance with the test procedures specified below. The tests shall be performed at 47, 51, and 57 Megahertz and the results shall comply with 3.10.13. Prior to the start of this test, the alignment indicator shall be OFF for at least 10 minutes. Measurements shall be made at each of the following temperatures:

a. 77°F

b. 32°F, 0°F, -40°F, and +140°F.

Test Procedures.-

a. Apply 12.0 volts from a d.c. supply to the plug in the alignment indicator internal battery compartment (upper left pin is positive polarity, lower right pin is negative polarity).

b. Set the FREQUENCY MC dial control to the desired position.

c. Connect the electronic counter to the RCVR ANTENNA socket by means of a small probe inserted in the socket.

d. Set the FUNCTION SWITCH to the RCVR ANT position.

e. Set the RCVR TEST SIG control in the extreme clockwise or number 6 position.

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f. Note and record the signal output frequency as read on the electronic counter.

g. The alignment indicator shall continue to operate for a 10 minute period.

h. Again note and record the signal output frequency as read on the electronic counter.

i. The difference in the frequency readings shall comply with 3.10.13.

4.9.14 Input Voltage Range (See 3.10.14).- To determine compliance with 3.10.14, the Signal Generator Output Test (4.9.10) shall be conducted at input voltages to the alignment indicator of 10.0 and 16.5 volts d.c.

4.9.15 Oscillator Operation (See 3.10.15).- To determine compliance with 3.10.15, the Signal Generator Output Test (4.9.10) shall be conducted at an input voltage of 9.0 volts d.c.

4.10 500 Hour Operation Test (See 3.11).-

4.10.1 Test Conditions.- The channel alignment indicator shall be operated continuously for 500 hours under the following conditions:

Standard Loads	The standard transmitter 50 ohm non-inductive load resistor used in test procedure 4.9.5 will be plugged into the XMTR ALIGN socket. The standard receiver load used in test procedure 4.9.7 shall be plugged into the receiver connector assembly block.
Duty Cycle	The indicator function switch shall be operated in sequence from extreme counter-clockwise to extreme clockwise position. One function switch rotation cycle shall be completed during a 10 minute operating period followed by a 50 minute OFF period with function switch in extreme counter-clockwise position. During the 50 minute OFF period of the function switch, the Battery Test switch shall be operated for one minute in the RCVR position and one minute in the XMTR position.

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Input Voltage (Internal battery)	Nominal voltage 50 per cent of the time. High voltage 25 per cent of the time. Low voltage 25 per cent of the time.
Input Voltage (Receiver battery)	Six volts during CYCLE time the battery test switch is in the RCVR position.
Input Voltage (Transmitter battery)	Sixteen volts during CYCLE time the battery test switch is in the XMTR position.
Vibration	10 minutes of each hour at 25 ± 5 Hz at 1/32 inch maximum amplitude.
Temperature	With test conditions, paragraph 3, General Requirements, MIL-STD-810.

4.10.2 Measurements.- The measurements listed below shall be made daily. Measurements shall be taken so that all tests are recorded under all conditions of input voltage and vibration during each 144 hours of tests.

Measurement	Inspection Para
Internal Battery	4.9.3
Transmitter Current	4.9.5
Receiver Oscillator Current	4.9.7
Signal Generator Output	4.9.10

4.11 Operational Inspection (See 3.15).- Immediately prior to packageing, the following tests shall be conducted:

a. Internal Battery Test.-

b. Field Strength Meter Test.- Use an AN/PRT-4( ) transmitter to radiate a signal with the alignment indicator function switch in XMTR ANT position. Insert the alignment indicator test antenna in the XMTR ANT socket, key the transmitter and observe whether the alignment indicator meter reads.

c. Alignment Indicator Internal Generator Signal Output Test.- Use an AN/PRR-9( ) receiver. Set the alignment indicator function switch to RCVR ANT position and set the RCVR TEST SIG control to clockwise position. Insert the test antenna in the RCVR ANT socket. Turn on the receiver until the squelch is off and there is noise output from the receiver horn. Set the alignment indicator FREQUENCY MC dial to the receiver frequency as determined by receiver quieting.

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4.12 Acceptance Inspection of Preparation for Delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.13 Rough Handling Test (Preparation for Delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract (6.2), the operational inspection (see 4.12) of the channel alignment indicator shall be conducted to determine freedom from operational malfunction caused by rough handling.

4.14 Visual and Mechanical Inspection.- The units shall be examined for the applicable defects listed in Standard MIL-STD-252.

4.15 Air-Seal Test.- The equipment shall be opened and closed again in such manner as to break and remake the seal. Immediately thereafter, the equipment as field transported shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the transit case or to the interior of the equipment inclosure when no transit case is provided. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. During this 1 minute period the decrease in vacuum shall not exceed 0.01 pound. The gage used for measurement of the vacuum shall be of such accuracy that a difference of 0.01 pound can be determined readily. (See 3.14)

4.16 Fungus.- The equipment (including accessories) shall be tested in accordance with Method 508, Procedure 1, of MIL-STD-810. Spraying shall include parts and materials on and under the chassis, and the interior of the equipment case. The equipment shall remain in an open condition during the test. Operation is not required. (See 3.9.8)

4.17 Salt Fog.- The equipment (including accessories) in its normal operating position shall be tested in accordance with Method 509, Procedure 1 of MIL-STD-810. The equipment shall not be opened. Operation is not required. No rust shall be imposed on the test item by the test facility. (See 3.9.9)

4.18 Electromagnetic Interference Characteristics (See 3.12).- The equipment shall be subjected to test methods CE06, RE02, CRE01, and RS03 as specified in MIL-STD-462. The specified test methods shall include the following spectrum requirements.

a. CE06, conducted Emission, Antenna Terminal, 10 KHz to 3 GHz.

b. RE02, Radiated Electric Field, Broadband, 14 KHz to 1 GHz.

c. CRE01, Transient Measurements, Manual Switching, 14 KHz to 1 GHz (Radiated).

d. RS03, Radiated Susceptibility, Electric Field, 14 KHz to 1 GHz.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.- Preservation and packaging shall be level A or C, as specified. (See 6.2c)

5.1.1 Level A.-

5.1.1.1 Cleaning.- Indicator, Channel Alignment ID-1189( )/PR shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying.- Indicator, Channel Alignment ID-1189( )/PR shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation Application.- None required.

5.1.1.4 Unit Packaging.- Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical Literature.- Each technical literature shall be packaged method 1C-1.

5.1.1.4.2 Indicator, Channel Alignment ID-1189( )/PR.- Each indicator shall be individually packaged Method III as follows: Accessories and hardware shall be securely placed within the mounting clips provided in the case. Close the case cover and secure all fastenings. Cushion each indicator on all surfaces with cells or pads, or both, fabricated of fiberboard conforming to PPP-F-320, grade W6c, designed to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned set within a close-fitting box conforming to PPP-B-636, W6c. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level C.- Each Indicator, Channel Alignment ID-1189( )/PR shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing.- Packing shall be level A, B, or C as specified. Shipping containers for all levels shall be capable of stocking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents. (See 6.1)

5.2.1 Level A.-

5.2.1.1 Consolidation.- A quantity of Indicator, Channel Alignment ID-1189( )/PR, packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. To facilitate palletization, fiberboard boxes shall be uniform in size and contain equal quantities of the packaged items to the greatest extent practicable.

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5.2.1.2 Palletized Load.- A quantity of containers, packed as specified in 5.2.1.1, shall be placed on a pallet, load type I conforming to MIL-STD-147, except that the pallet shall be softened conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having the sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.3 Less than Palletized Load.- When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.1 shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, class B.

5.2.2 Level B.-

5.2.2.1 Consolidation.- A quantity of Indicator, Channel Alignment ID-1189( )/PR, packaged as specified in 5.1, shall be packaged as specified in 5.2.1.1.

5.2.2.2 Palletized Load.- A quantity of containers, packed as specified in 5.2.2.1 shall be palletized as specified in 5.2.1.2.

5.2.2.3 Less than Palletized Load.- When quantities per destination are less than a pallet load, containers specified in 5.2.2.1 shall be reinforced by pressure-sensitive filament tape conforming to PPP-T-97, type IV as specified in the appendix to the box specification. No further packing shall be required.

5.2.3 Level C.-

5.2.3.1 Consolidation.- A quantity of Indicator, Channel Alignment ID-1189( )/PR, packaged as specified in 5.1, shall be packed as specified in 5.2.1.1, except that the fiberboard boxes shall be class domestic.

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5.2.3.2 Palletized Load.- A quantity of containers, packed as specified in 5.2.3.1, shall be palletized as specified in 5.2.1.2, except that the fiberboard cap shall be class domestic.

5.2.3.3 Less than Palletized Load.- When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.3.1 shall be used as the shipping container. No further packing shall be required.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use.- The Channel Alignment Indicator ID-1189( )/PR is intended to provide a portable, self-contained equipment to align Radio Receiver AN/PRR-9( ) and Radio Transmitter AN/PRT-4( ) on any operating channel in the 47 to 57 Megahertz range.

6.2 Ordering Data.- Procurement documents should specify the following:

a. Title, number and data of this specification and any amendment thereto.

b. Type required.

c. Levels of preservation and packaging, and packing. (See Sec. 5)

d. The specific paragraphs of Section 5 which are applicable to the particular procurement.

e. First Articles Inspection - Five sample units of Indicator, Channel Alignment are required so that lengthy environmental tests can be completed on one sample unit while complete performance measurements can be made on the other sample units. (See 3.1)

f. Marking and shipping of samples.

g. Place of final inspection.

h. Technical literature required. (3.16)

i. Whether rough handling and functional tests are required.

j. Dimensional Data - Dimensions of packaging materials in bill of materials table based on the dimensions of the equipment cited in the table. When the dimensions of the equipment vary, the dimensions of packaging materials shall be adjusted accordingly.

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6.3 Color.- The color chip furnished by the procuring agency (3.6) will be color chip number X-24087, and may be obtained upon request to Commander, US Navy Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 Group C Inspection.- Approval to ship may be withheld, at the discretion of the government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2)

6.5 Location of Operational Inspection and Air-Seal Test.- It is desirable that the operational inspection (4.11) and the air-seal test (4.15) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the air-seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.6 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: ID-1189W/PR. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.7 Verification Inspection.- Verification by the government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:
ARMY - EL

Preparing Activity:
ARMY - EL
Project No. 5820-A591

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-I-55510B(EL) INDICATOR, CHANNEL ALIGNMENT ID-1189( )/PR

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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